Hubbard-Holstein report — L=2

Parameter manifest

Experiment: Prepared-state quality and exact-vs-propagated dynamics for Hubbard-Holstein using hh_hva_ptw state preparation and CFQM4 evolution.

Model and regime:
- Model family: Hubbard-Holstein
- Problem: hh
- L: 2
- Boundary: open
- Ordering: blocked
- Drive enabled: true

Method chain:
- Ansatz type: hh_hva_ptw
- Optimizer: SPSA
- Internal VQE method: hardcoded_hh_hva_physical_termwise_statevector
- Propagator: CFQM4
- Initial state source: adapt_json
- Selected propagated branch: PAOP seed
- Branch order: Exact filtered GS, Exact PAOP, Trotter PAOP, Exact HVA, Trotter HVA

Core physical parameters:
- t: 1
- U: 4
- dv: 0

Trajectory settings:
- t_final: 10
- num_times: 135
- trotter_steps: 128
- Suzuki order: 2

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  - Term order: sorted

SPSA settings:
  - a: 0.2
  - c: 0.1
  - A: 10
  - alpha: 0.602
  - gamma: 0.101
  - eval repeats: 1
  - eval aggregate: mean
  - average last: 0

Drive settings:
  - A: 0.5
  - omega: 2
  - tbar: 3
  - phi: 0
  - t0: 0
  - Pattern: staggered
  - Time sampling: midpoint
  - Exact-steps multiplier: 2

Hubbard-Holstein parameters:
  - omega0: 1
  - g_ep: 0.5
  - n_ph_max: 1
  - Boson encoding: binary
  - nq_fermion: 4
  - nq_phonon: 2
  - nq_total: 6

Branch provenance:
  - PAOP source: adapt_json
  - PAOP pool: full_meta
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  - PAOP depth: n/a
  - HVA source: regular_vqe
  - HVA ansatz: hh_hva_ptw
  - HVA VQE success: true

Imported ADAPT state:
  - Source JSON: L2_hh_smart_replay_bundle_diagnostic.json
  - Metadata match passed: true
  - Strict match: true
  - Pool: full_meta
  - Ansatz depth: n/a
  - ADAPT energy: 0.168226029962
  - ADAPT |ΔE|: 0.00955812583648

Notes:
  - The parameter manifest is intentionally separate from the executed command; full provenance appears in the
    appendix.
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Executive summary

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Experiment: Scientist-facing overview of the selected paop seed versus exact reference.

Headline results:
- Run mode: drive-enabled
- Subspace fidelity at t=0: 0.996992587824
- VQE energy: 0.179738578063
- Exact filtered energy: 0.158667904126
- |VQE - exact(filtered)|: 0.0210706739374
- QPE energy estimate: n/a

Prepared state and propagation:
- Ansatz: hh_hva_ptw
- Propagator: CFQM4
- Initial state source: adapt_json
- Selected propagated branch: PAOP seed
- Filtered sector: {"n_dn": 1, "n_up": 1}

Trajectory coverage:
- t_final: 10
- num_times: 135
- trotter_steps: 128
- Boundary: open
- Ordering: blocked

Notes:
- Core result pages focus on prepared-state quality and exact-versus-propagated observables.
- 3D surfaces, branch-complete diagnostics, imported-state metadata, and the full command are appendix material.

Core scientific results

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Prepared-state quality is shown first; supporting diagnostic surfaces are deferred to the appendix.

Focus:
- Reference / prepared / propagated observables only.
- Drive diagnostics stay near the main dynamics pages when drive is enabled.
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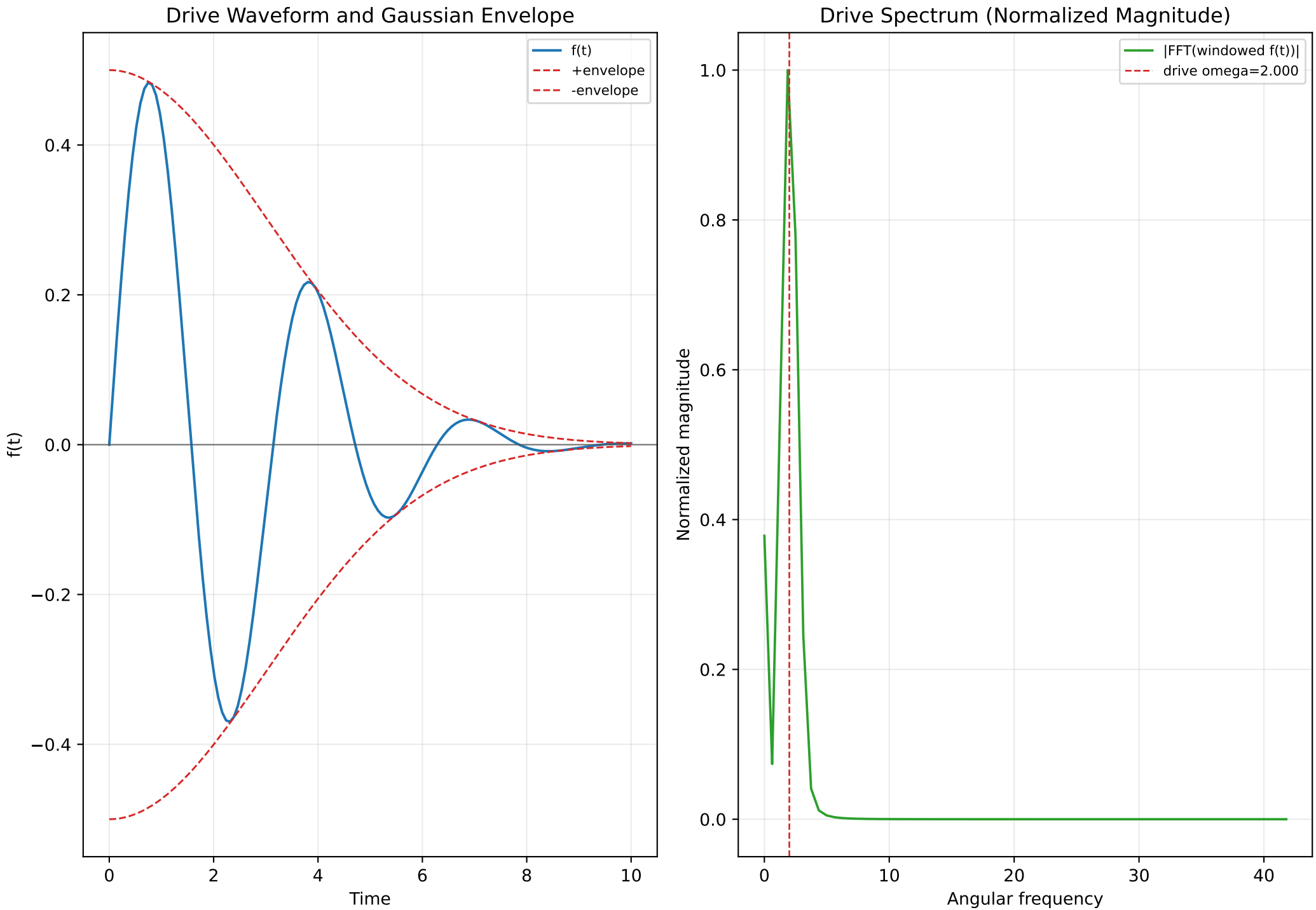
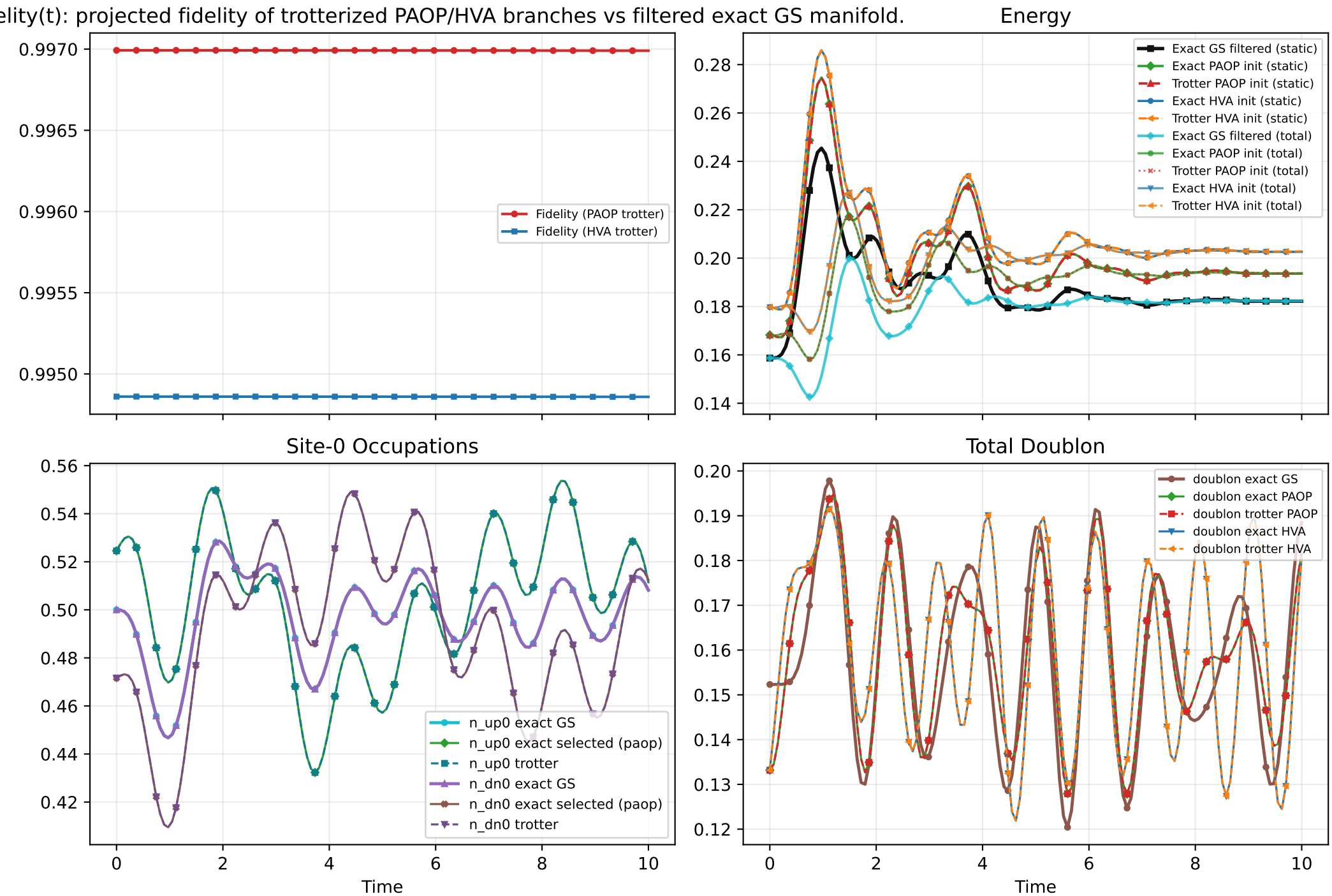
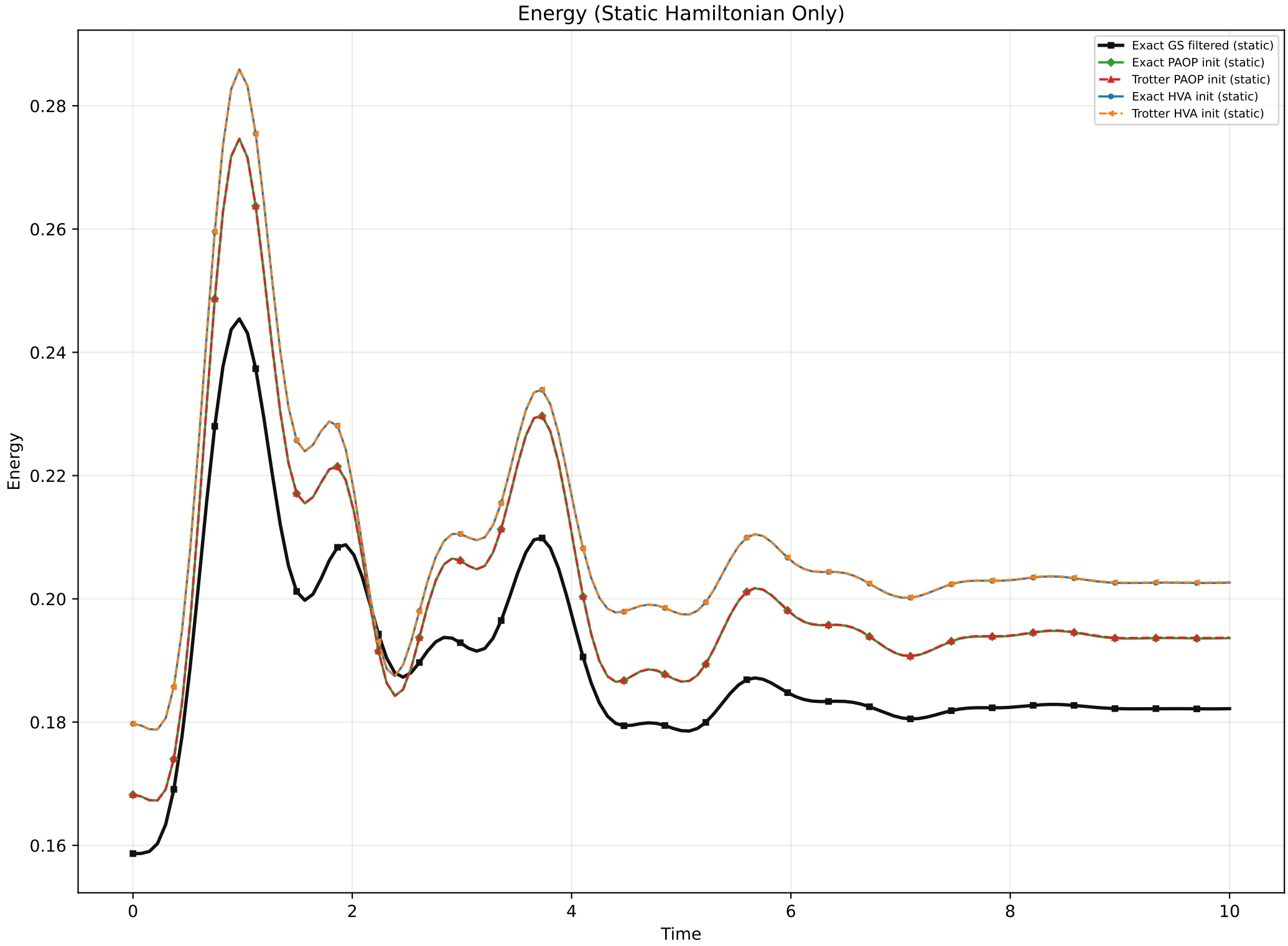
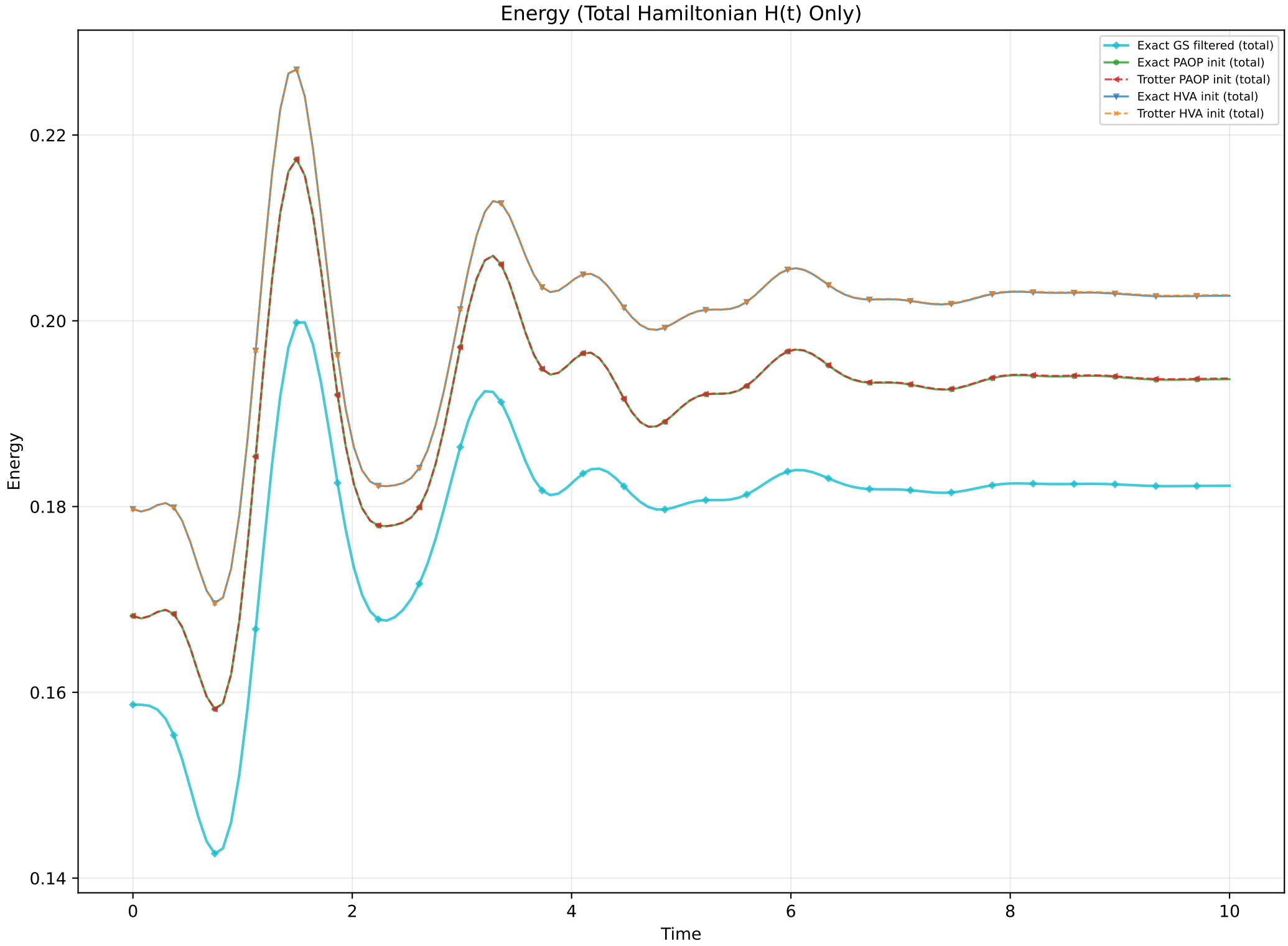
Drive Diagnostics: A=0.5, omega=2.0, tbar=3.0, phi=0.0, t0=0.0



Hardcoded Hubbard Pipeline: L=2




Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)
Energy
0.28
0.26
0.24
0.22
0.20
0.18
0.16
0
2
4
6
8
10
Time


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact PAOP init (total)
Trotter PAOP init (total)
Exact HVA init (total)
Trotter HVA init (total)
Energy
0.22
0.20
0.18
0.16
0.14
0
2
4
6
8
10
Time

Technical appendix

```
Supporting diagnostics, imported-state provenance, branch-complete views, and reproducibility material.

Focus:
- 3D density surfaces and scalar lane plots.
- Absolute-error heatmaps.
- Imported ADAPT state metadata when present.
- Full executed command for reproducibility.
```

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ADAPT Initial-State Provenance

import_json: L2_hh_smart_replay_bundle_diagnostic.json
import_source: replay_vqe
strict_match: True
metadata_match_passed: True

pool_type: full_meta
ansatz_depth: None
num_parameters: None
operator_count: None
adapt_energy: 0.16822602996220284
adapt_abs_delta_e: 0.0095581258364765

Dynamics used imported ADAPT state as t=0 initial condition.
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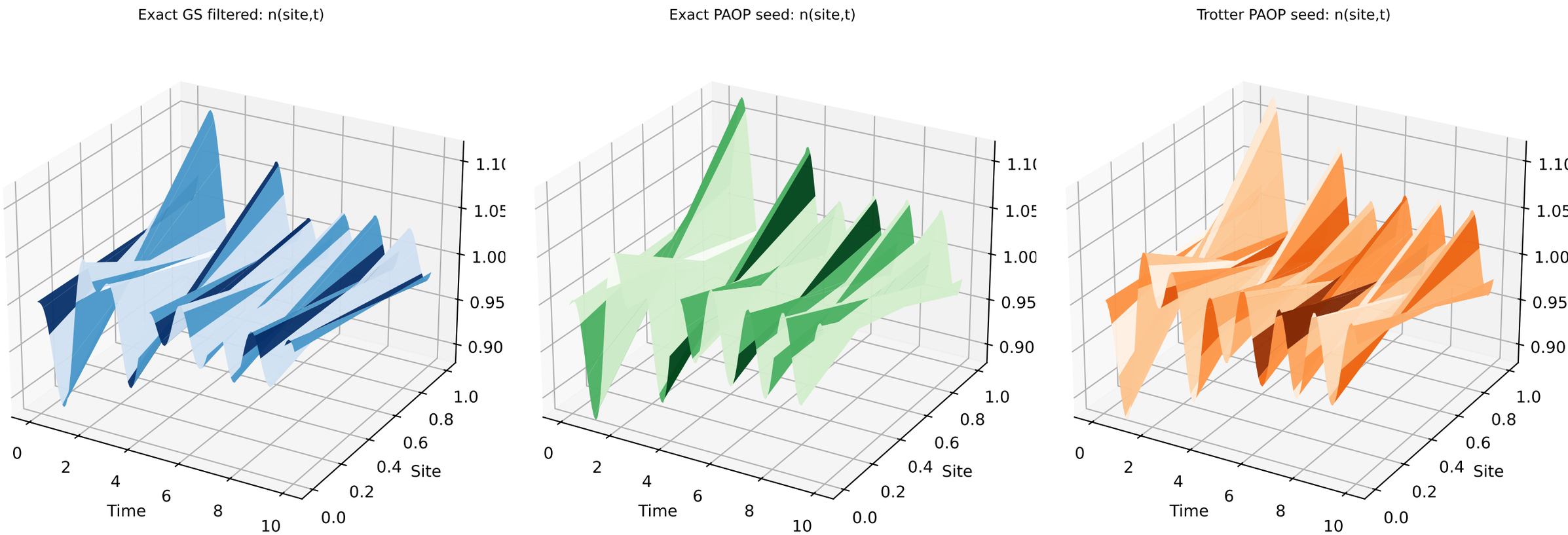
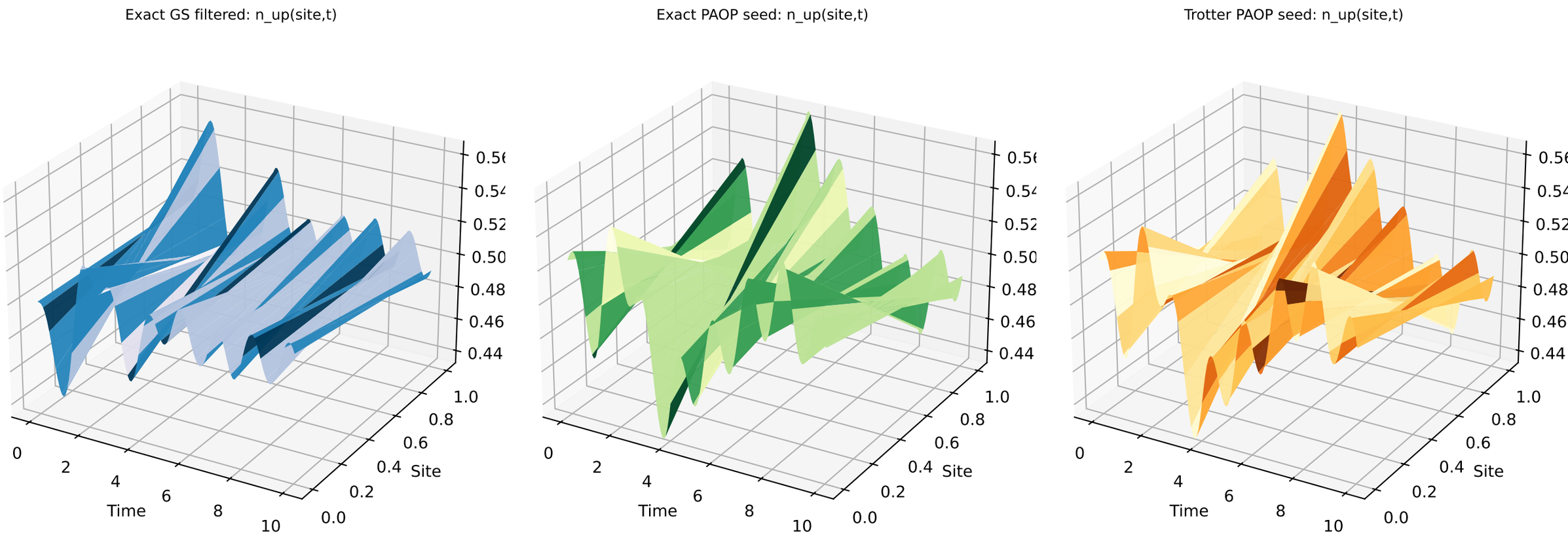
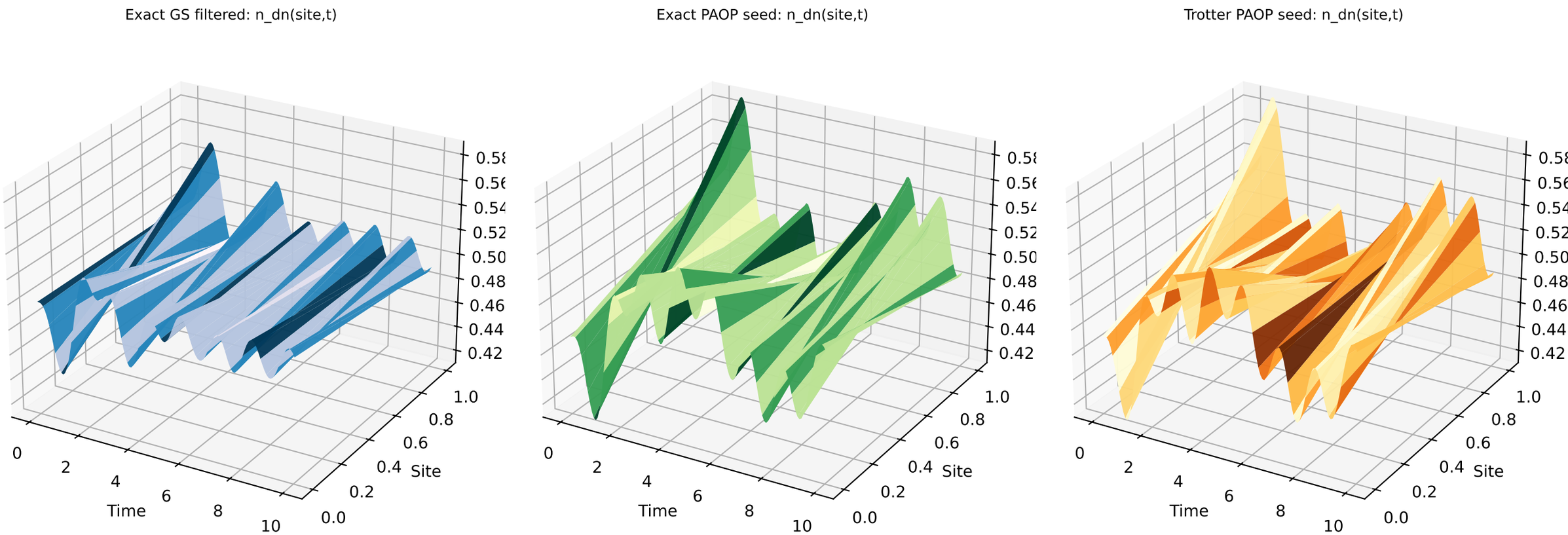
L=2 3D Densities (Total n)



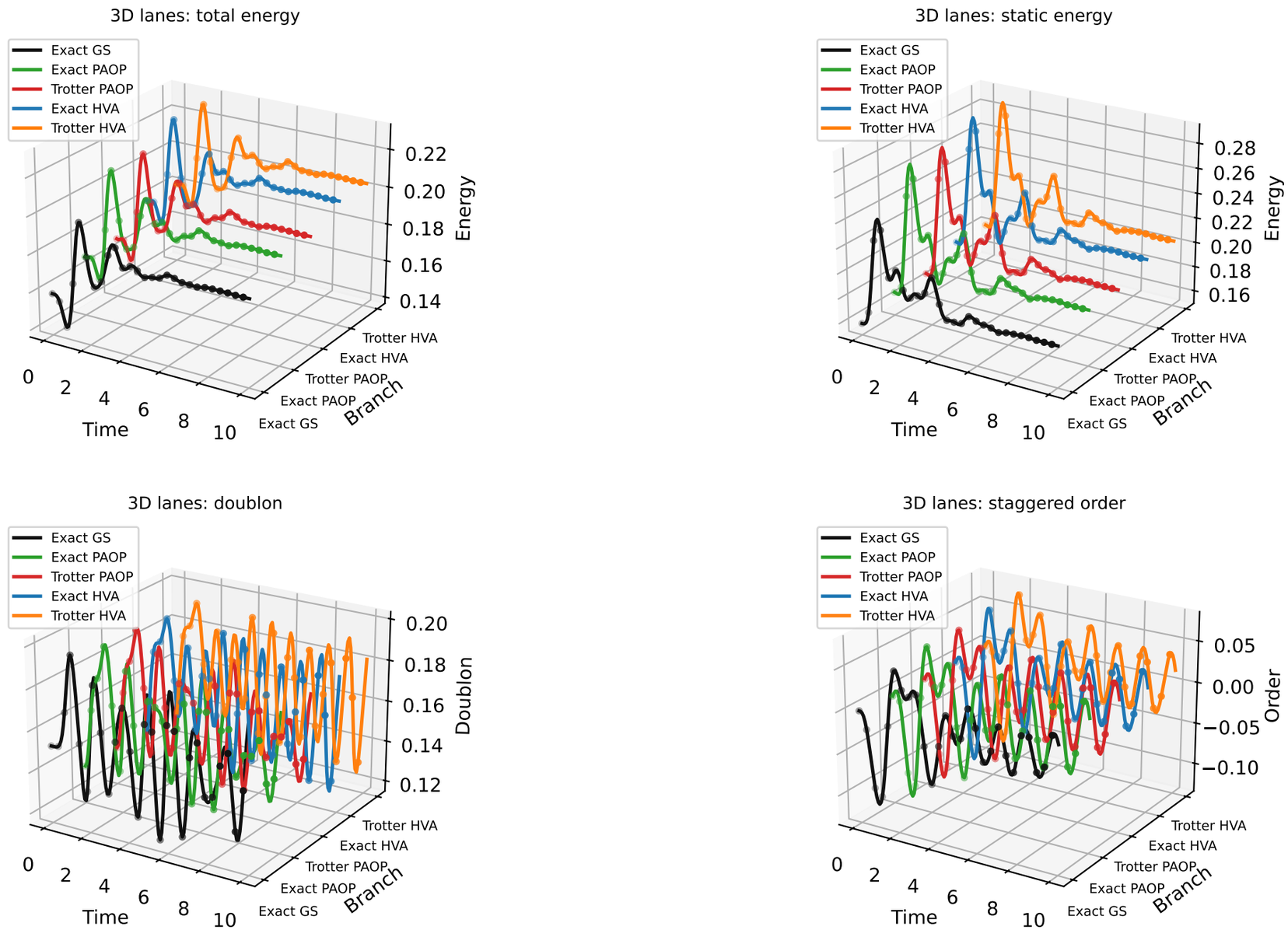
L=2 3D Densities (Spin-Up)

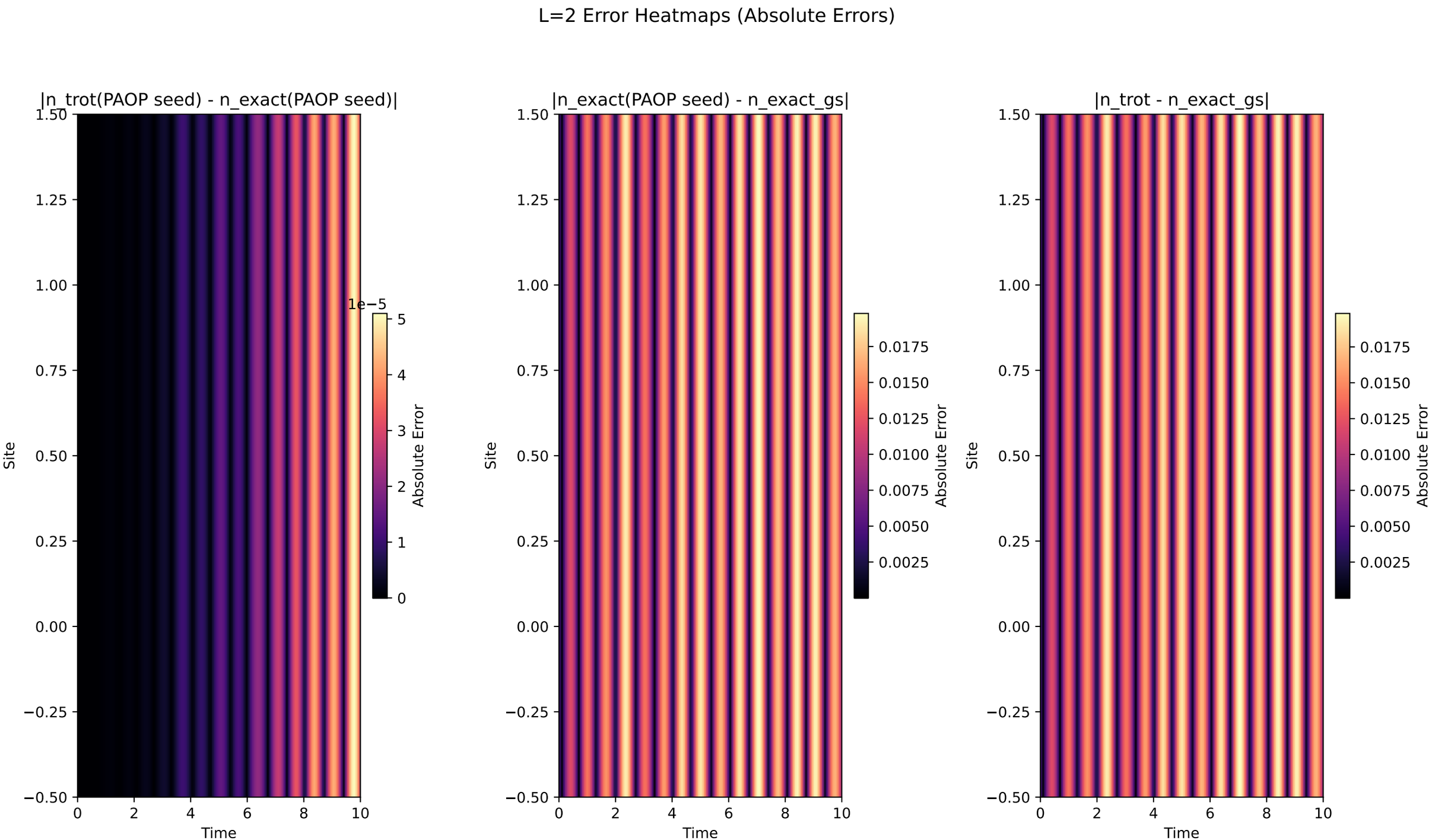


L=2 3D Densities (Spin-Down)

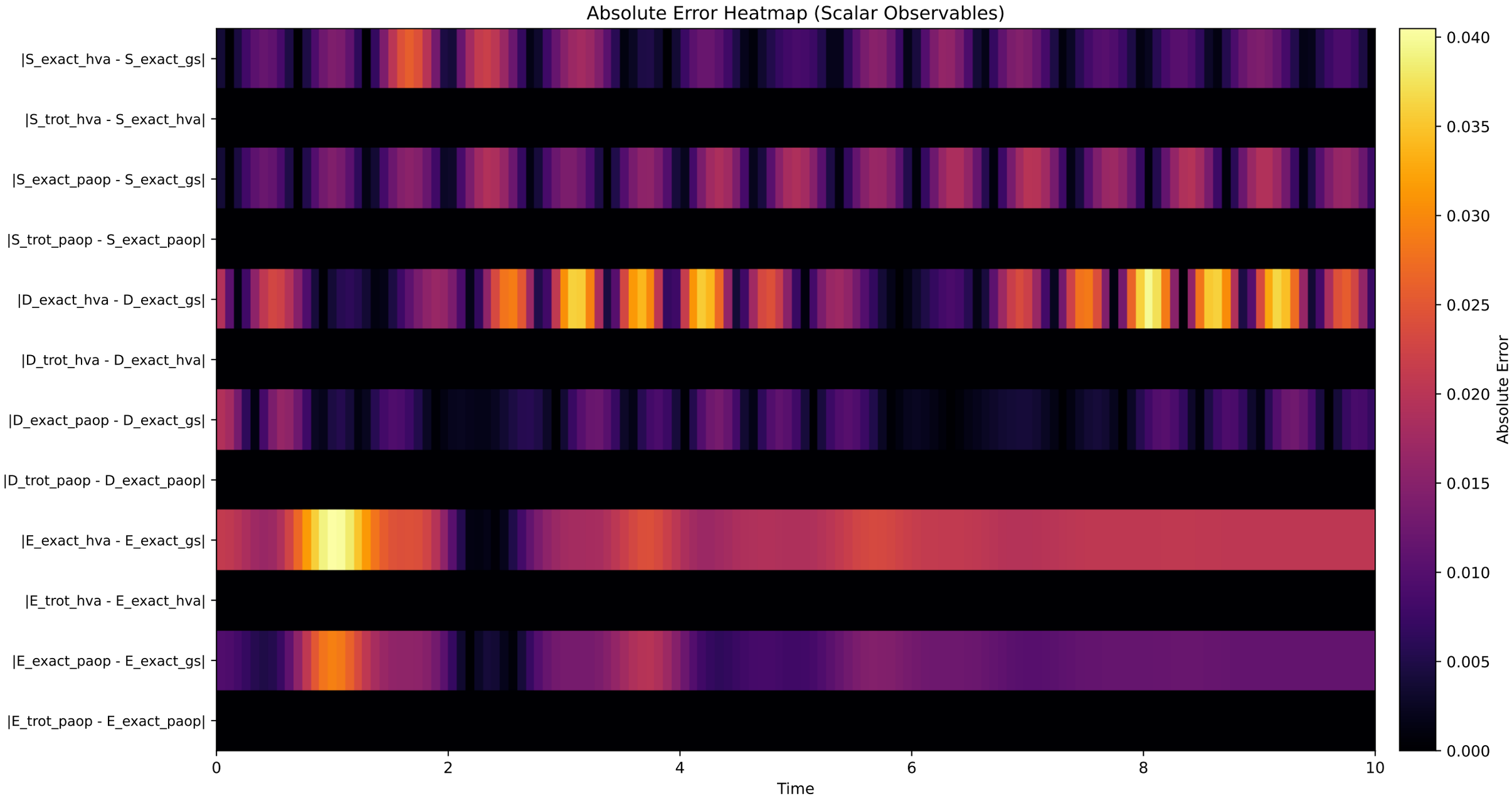


L=2 3D Scalar Observables (Five Branches)

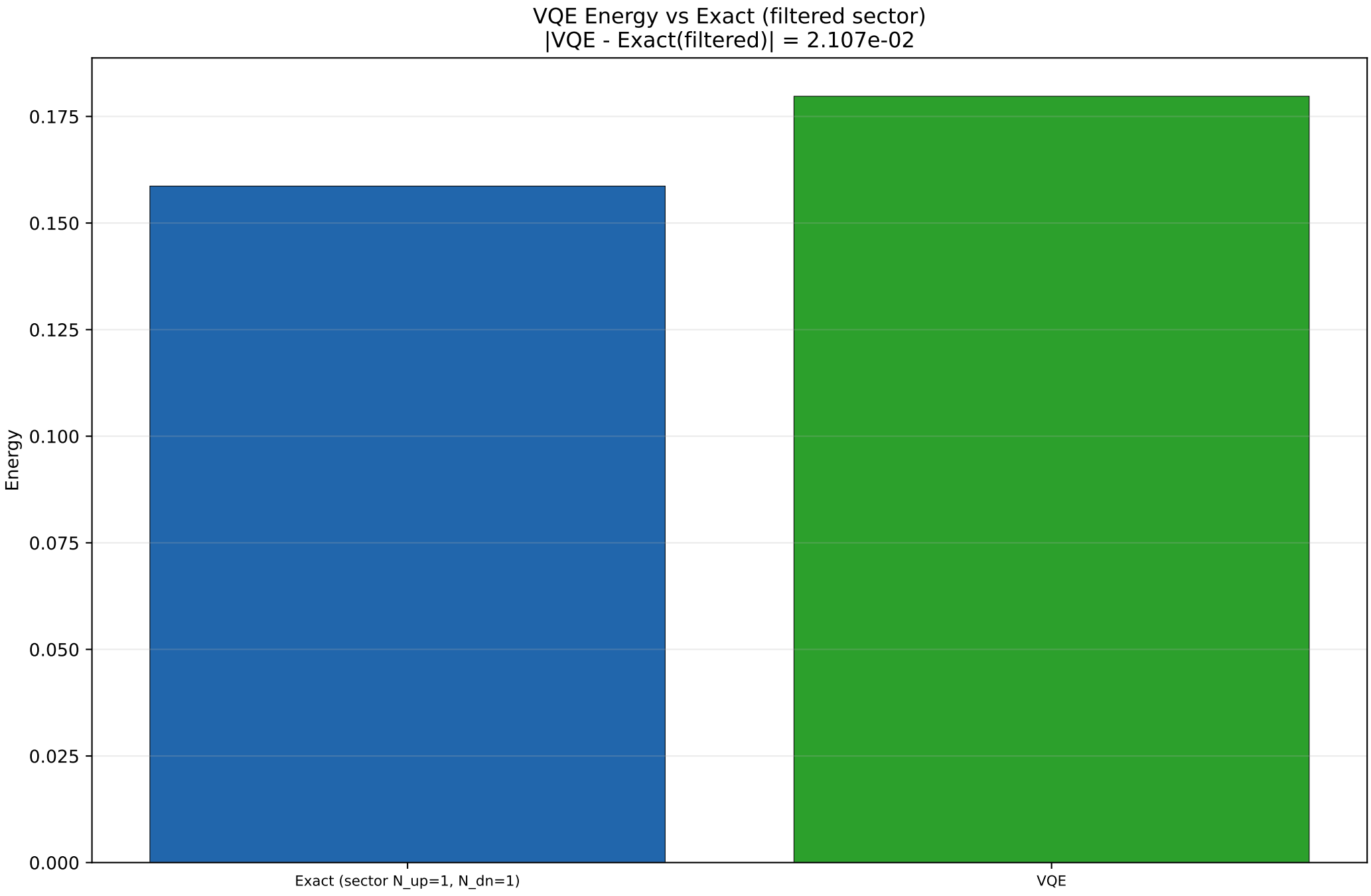



L=2 Error Heatmaps (Absolute Errors)
|n_trot(PAOP seed) - n_exact(PAOP seed)|
|n_exact(PAOP seed) - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1e−5
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
2
4
6
8
10
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025

L=2 Scalar Error Heatmap



VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva_ptw
  - vqe method: hardcoded_hh_hva_physical_termwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.996992587824234
  - subspace_fidelity_paop_t0: 0.996992587824234
  - subspace_fidelity_hva_t0: 0.9948605328409531
  - energy_t0_exact_gs: 0.15866790412572584
  - energy_t0_exact_paop: 0.16822602996220287
  - energy_t0_trotter_paop: 0.16822602996220287
  - energy_t0_exact_hva: 0.17973857806309487
  - energy_t0_trotter_hva: 0.17973857806309487
  - ground_state_exact_energy_full_hilbert: -0.150680123886
  - ground_state_exact_energy_filtered: 0.15866790412572704
  - filtered_sector: {'n_up': 1, 'n_dn': 1}
  - vqe_energy: 0.1797385780630951
  - qpe_energy_estimate: None

Config:
  - initial_state_source: adapt_json
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity_paop_trotter(t) = <psi_paop_trot(t)|P_exact_gs_subspace(t)|psi_paop_trot(t)> and
    fidelity_hva_trotter(t) = <psi_hva_trot(t)|P_exact_gs_subspace(t)|psi_hva_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol. Legacy key fidelity(t) follows the selected initial-state branch.
  - hamiltonian_terms: 17
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```

Executed Command

Reference: pipelines/run_guide.md
Script: pipelines/hardcoded/hubbard_pipeline.py

```
/opt/anaconda3/bin/python pipelines/hardcoded/hubbard_pipeline.py --L 2 --problem hh --t 1.0 --u 4.0 --dv 0.0
    --omega0 1.0 --g-ep 0.5 --n-ph-max 1 --boson-encoding binary --boundary open --ordering blocked --initial-
    state-source adapt_json --adapt-input-json L2_hh_smart_replay_bundle_diagnostic.json --enable-drive
    --drive-A 0.5 --drive-omega 2.0 --drive-tbar 3.0 --drive-phi 0.0 --drive-pattern staggered --drive-time-
    sampling midpoint --t-final 10.0 --num-times 135 --trotter-steps 128 --exact-steps-multiplier 2 --vqe-
    ansatz hh_hva_ptw --vqe-reps 2 --vqe-restarts 4 --vqe-maxiter 1778 --vqe-method SPSA --vqe-seed 7 --skip-
    qpe --output-json L2_hh_smart_results_diagnostic.json --output-pdf L2_hh_smart_results_diagnostic.pdf
```